PROJECT PROPOSAL

Title: India House Price

1. Student Details -

- **Name :** Kasture Swarada Pramod
- **E-mail :** swaradakasture0206@gmail.com
- **Class :** T.Y. ECM
- **Roll Number :** 33

---

2. Problem Statement -

The real estate sector, house prices vary due to factors such as location, area, number of bedrooms, amenities, and market demand. It becomes difficult for buyers, sellers, and investors to understand property pricing trends.

The aim of this project is to analyze house price data in India to understand price trends, identify factors affecting house prices, and generate useful insights for better decision-making.

---

3. Domain -

Real Estate / E-commerce / Data Analytics

This project belongs to the **Real Estate** domain because it deals with analysis of residential property prices in different locations across India. It focuses on studying factors such as location, area, number of bedrooms, amenities, and other property features that affect house prices.

It also comes under **E-commerce** because online property platforms like MagicBricks, 99acres, and Housing.com provide digital property listings, making buying, selling, and renting possible through online systems.

The project is related to **Data Analytics** because it uses data preprocessing, exploratory data analysis (EDA), visualization, and statistical methods to find patterns, trends, and useful insights from the dataset. Data analytics techniques help in understanding pricing behavior and supporting decision-making in the real estate sector.

---

4. Dataset Description –

- **Source: Kaggle**
- **Dataset Name: House Price Dataset - India**
- **Link: House Price Dataset - India (Kaggle)**

(https://www.kaggle.com/datasets/saincoder404/india-house-prices)

Dataset Features:
The dataset contains the following information:

- Location of property
- House price
- Area (Square Feet)
- Number of bedrooms (BHK)
- Number of bathrooms
- Furnishing status
- Property type
- Amenities and facilities

This dataset is useful for analyzing house price trends and factors affecting property prices.

---

5. Objectives

- To analyze house price trends in India
- To identify factors affecting house prices
- To study relationship between area and price
- To visualize data using graphs and charts
- To generate useful insights for decision-making

---

6. Methodology

The project will follow the following methodology:

1. Data Collection

The India House Price dataset will be collected from Kaggle. The dataset contains information related to house prices, location, area, number of bedrooms, and other property features required for analysis.

2. Data Cleaning and Preprocessing

The collected data will be cleaned by handling missing values, removing duplicate records, correcting inconsistent data, and converting data into proper formats for analysis.

3. Exploratory Data Analysis (EDA)

The dataset will be analyzed to understand price distributions, identify trends, compare prices across locations, and study relationships between important variables.

4. Data Visualization

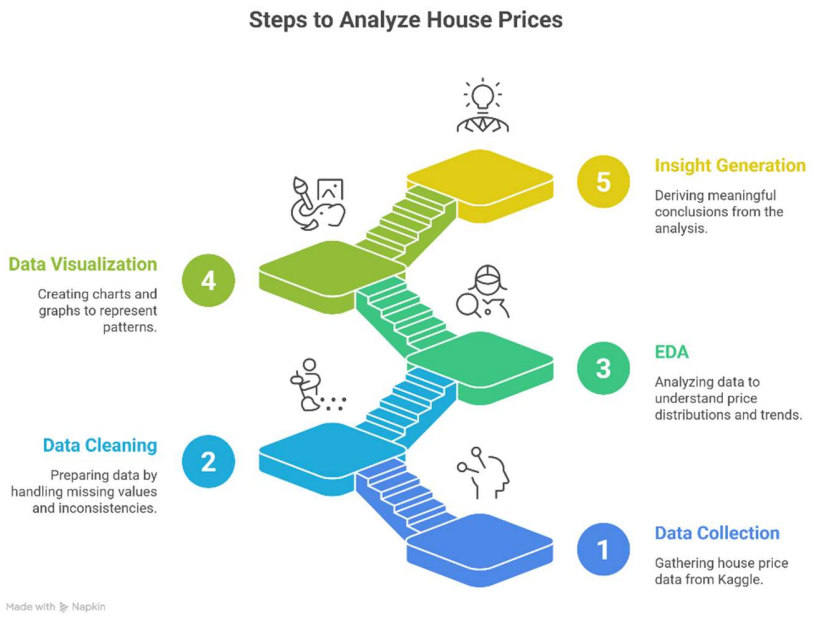
Different visualization techniques such as bar charts, histograms, scatter plots, box plots, and heatmaps will be used to represent patterns and trends clearly.

5. Insight Generation

Important insights will be generated from the analysis to identify major factors affecting house prices and support decision-making.

6. Tools and Technologies Used

The project will be implemented using Python, Pandas, Matplotlib, Seaborn, and Jupyter Notebook for data analysis and visualization.



---

5. Tools & Technologies

- **Python** – For data analysis and programming
- **NumPy** – For numerical computations
- **Plotly** – For interactive data visualization
- **Scikit-learn** – For basic predictive analysis
- **Google Colab** – For coding and project execution
- **Power BI** – For dashboards and visualization
- **Excel** – For basic data cleaning and analysis
- **Google Colab / Jupyter Notebook** – For implementation

---

7. Expected Outcome

- Comprehensive understanding of house price behavior and market trends in India
- Identification of key parameters such as location, area, and amenities influencing property prices
- Comparative analysis of house prices across different cities and regions
- Discovery of hidden patterns, correlations, and trends through exploratory data analysis
- Effective visual representation of data using charts and graphs for better interpretation
- Generation of meaningful insights to support buyers, sellers, and investors in decision-making.

---

8. Conclusion

This project will help analyze house price trends in India and identify factors affecting property prices using data analytics techniques. The insights generated from the analysis can support better decision-making for buyers, sellers, and investors.